

Cardiovascular system

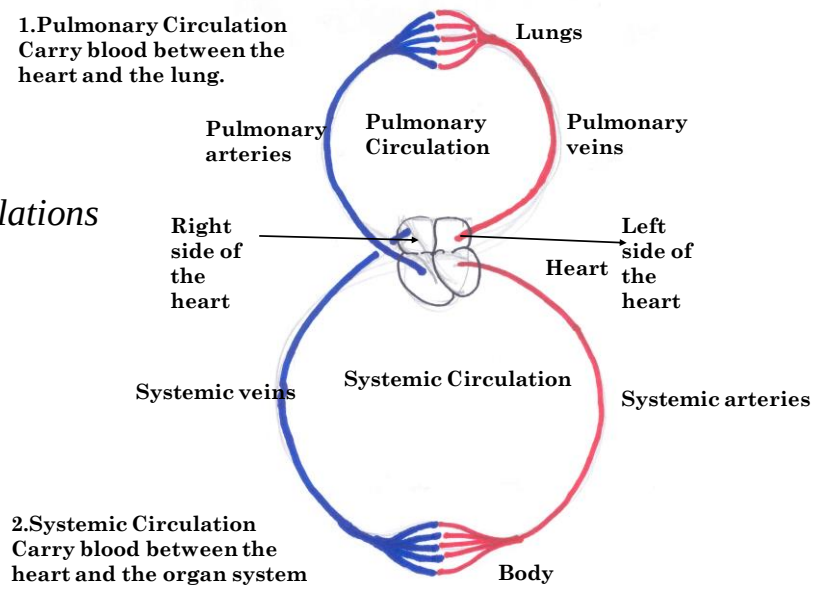
- The **circulatory system** is the transport system that supplies O₂ and substances absorbed from the gastrointestinal tract to the tissues, returns CO₂ to the lungs and products of the metabolism to the kidneys

It's divided into:-

A. major (systemic) circulation

B. lesser (pulmonary) circulation

The right figure showing these circulations



Cardiovascular System Functions:

1. Transport

- A. Dissolved gases: O₂ and CO₂
- B. Nutrients: Amino acids, carbohydrates, fatty acids, etc.
- C. Metabolic wastes: NH₃, urea, uric acid, etc.
- D. Enzymes/hormones
- E. Heat

2. Homeostatic

- A. pH and electrolyte regulation
- B. Heat regulation

3. Natural defense

- A. Clot formation
- B. Immunity
 - 1. Short term/phagocytosis
 - 2. Long term/antibody-antigen reactions

The circulatory system consists of three basic components:

- 1. Heart
- 2. Blood Vessels
- 3. Blood

Heart Functions:

1. A muscular pump which facilitates optimal blood flow to the body.
2. Interconnects the pulmonary and systemic circulation.
3. Segregates oxygenated/deoxygenated blood.
4. Maintains one-way blood flow.
5. Produces the hormone ANF (atrial natriuretic factor) which helps to regulate blood pressure.

Heart valves:

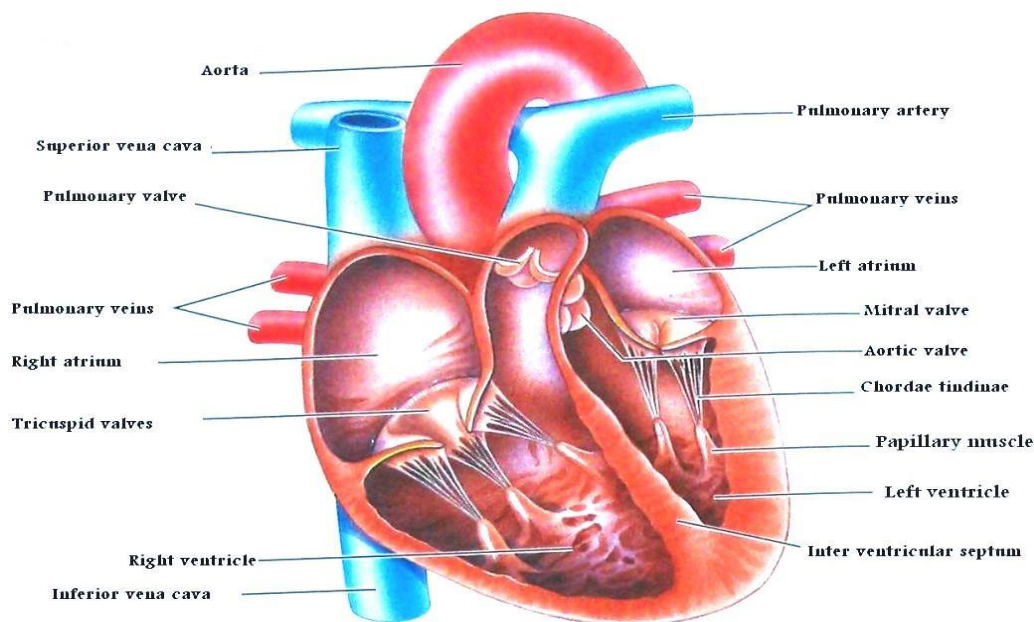
The heart valves include:

A. Atrioventricular Valves.

- The A-V valves (the tricuspid and mitral valves) prevent backflow of blood from the ventricles to the atria during systole,

B. the semilunar valves (the aortic and pulmonary artery valves) prevent backflow from the aorta and pulmonary arteries into the ventricles during diastole.

- During cardiac cycle heart valves open and close in response to differences in blood pressure on their two sides
- The following figure showing these valves and their locations



Circuit of Blood Flow

- The venae cavae are veins returning blood to the right atrium. Oxygen has been extracted from this blood. Carbon dioxide has been added to it. This blood is pumped from the right ventricle through the pulmonary artery to the lungs.
- The lungs add oxygen to this blood received from the right side of the heart. Carbon dioxide is removed from this blood. This blood flows through pulmonary veins to

the left atrium of the heart. This oxygen rich blood is pumped from the left ventricle through the aorta, a large artery.

Coronary Circulation

- Heart muscle is supplied with oxygen and nutrients by blood delivered to it by coronary circulation, not from blood within heart chambers
- Heart receives most of its own blood supply that occurs during diastole
- During systole, coronary vessels are compressed by contracting heart muscle
- Coronary blood flow normally varies to keep pace with cardiac oxygen needs.

Types of cardiac muscle fibers:

The heart is composed of three types of cardiac muscle fibers:

1-Atrial and ventricular types: which contract to some extent in the same manner as skeletal muscle fibers.

2-Rhythmic type: for the initiation of impulses (e.g.) sino atrial node (S.A.N) and atrio-ventricular node (A.V.N.).

3-Conductive type: contract feebly because they contain few contractile fibrils, but a greater content of sarcoplasm and glycogen, it is concerned for rapid conduction of impulses through the heart.

Properties of Cardiac muscle:

- *Rhythmicity*
- *Excitability*
- *Conductivity*
- *contractility*

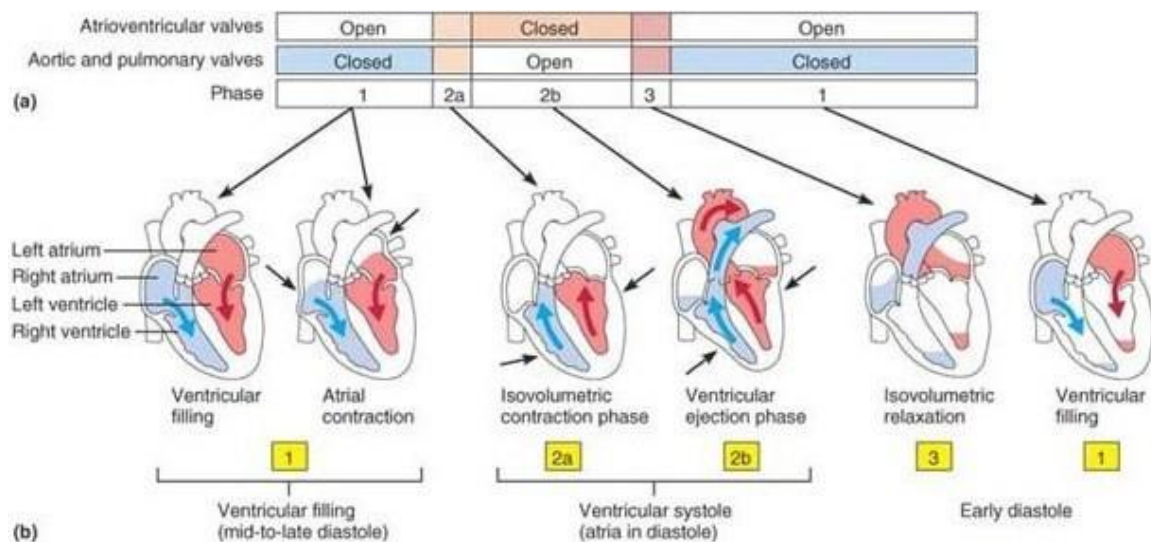
The Cardiac Cycle

Definition: The cardiac events that occur from the beginning of one heartbeat to the beginning of the next are called the cardiac cycle

- Each cycle is initiated by spontaneous generation of an action potential in the sinus node.
- The cardiac cycle consists of a period of relaxation called **diastole**, during which the heart fills with blood, followed by a period of contraction called **systole**.
- The normal number of cardiac cycles per minute ranges from 60 to 80.

Phases of cardiac cycle: divided into three principles phases

1. Ventricular filling (Mid to late ventricular diastole)
 - a. Blood flow passively into the atria through open AV valves and into ventricles (80%) where the pressure is lower, and the aortic and pulmonary valves are closed
 - b. Atria contract, forcing the remaining blood into ventricles (20%)
2. Ventricular systole divided into:
 - a. Isovolumetric contraction: ventricles contract and intraventricular pressure rises, closing AV valves. Briefly, ventricles completely closed chambers.
 - b. ventricular ejection : rising ventricular pressure forces semilunar valves open and blood ejected from the heart
3. Isovolumetric relaxation:
 - During this phase the ventricles relax and ventricular pressure drops.
 - Blood backflow, closing semilunar valves. Ventricles are totally closed off again. Meanwhile, the atria have been filling with blood. When atrial pressure exceeds ventricular pressure, AV valves open and ventricular filling, phase 1 begins again
 - The following figure showing these phases



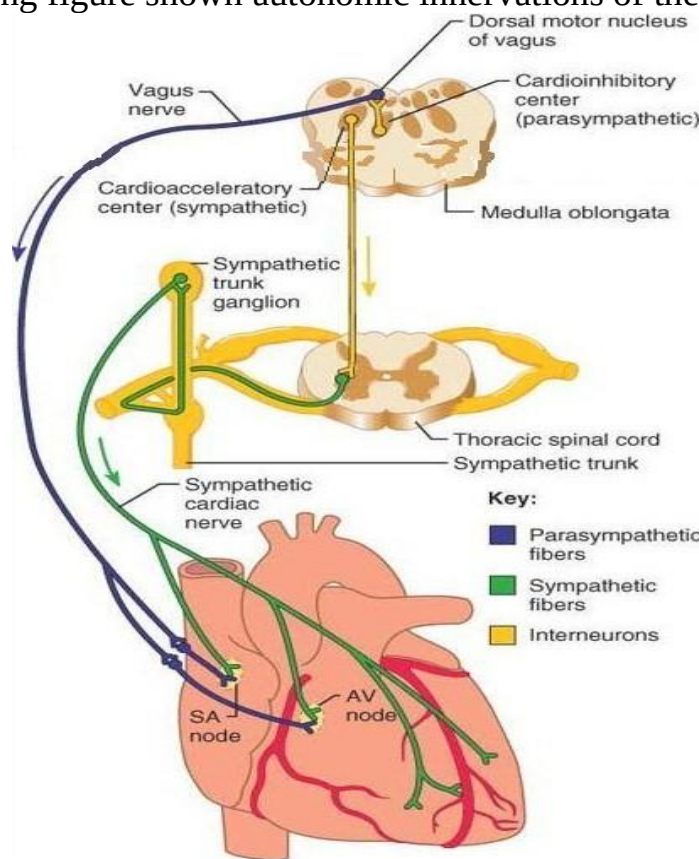
Blood flow controlled by pressure change:

- Pressure changes reflect the alternating contraction and relaxation of the heart
- Blood moves along a pressure gradient (from higher to lower pressure) through any available opening

- Pressure changes cause the heart valves to open and close, which keeps blood flowing in the forward direction

Conducting system of the heart:

- The heart has two controls as the following:
- a- External Controls :** by parasympathetic and sympathetic
- The heart is influenced by autonomic nerves originating in the **cardiovascular centre in the medulla oblongata** which reach it through the autonomic nervous system.
- The **vagus nerves** (parasympathetic) supply mainly the **SA and AV nodes and atrial muscle**.
- **Parasympathetic** decrease the rate and force of the heart beat.
- The sympathetic nerves supply **the SA and AV nodes and the myocardium of atria and ventricles**.
- **Sympathetic stimulation** increases the rate and force of the heart beat.
- The sympathetic and parasympathetic actions **are antagonistic to one another**.
- The following figure shown autonomic innervations of the heart

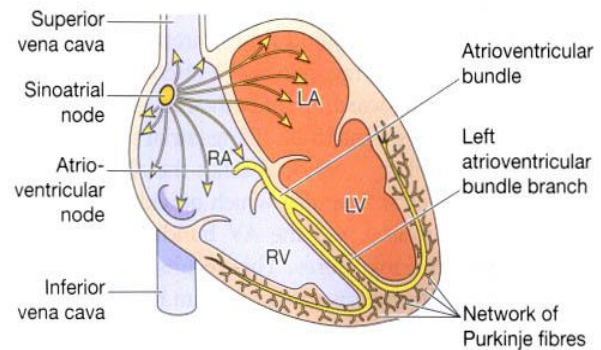
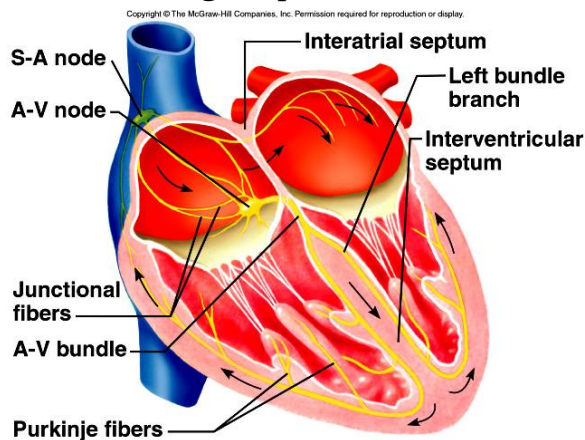


b- Internal Controls (intrinsic system)which involved:

1. SA node

2. AV node
3. Bundle of his
4. Right bundle
5. Left bundle
6. Purkinje fibers

- The heart has an *intrinsic system* whereby the cardiac muscle is automatically stimulated to contract without the need for a nerve supply from the brain
- The intrinsic system can be stimulated or depressed by nerve impulses initiated in the brain and by circulating chemicals including hormones.
- Small groups of specialized neuromuscular cells in the myocardium which initiate and conduct impulses causing coordinated and synchronized contraction of the heart muscle.
- The following shape shown the *intrinsic system* and their locations



Sinoatrial node (SA node)

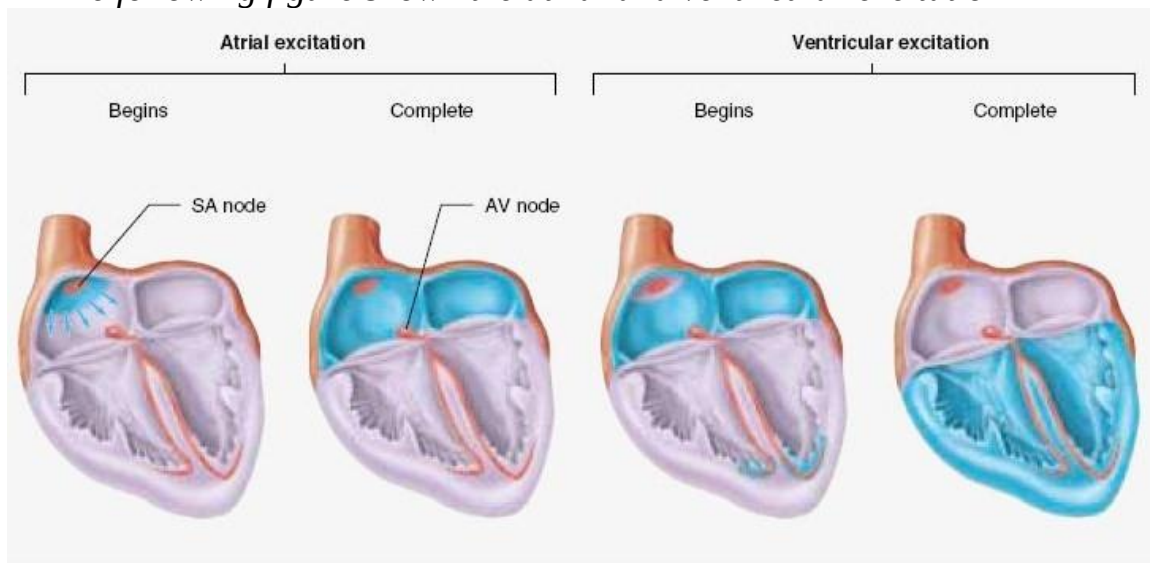
- Is a small mass of specialized cells is *in the wall of the right atrium* near the opening of the superior vena cava.
- SA node is called the '**pace-maker**' of the heart because it normally initiates impulses more rapidly than other groups of neuromuscular cells.
- *Function of SA node* initiates heart beat.

Atrioventricular node (AV node)

- **AV node** is a small mass of *neuromuscular tissue* which situated in the wall of the **atrial septum** near the atrioventricular valves.
- Normally the AV node is stimulated by impulses that sweep over the atrial myocardium.
- It is capable of initiating impulses that cause contraction but at a **slower rate than the SA node.**

Atrioventricular bundle (AV bundle or bundle of His)

- **AV bundle** is a mass of specialised fibres that originate from the AV node.
- The AV bundle crosses the fibrous ring that separates atria and ventricles then, at the upper end of the ventricular septum; it divides into right and left bundle branches.
- Within the ventricular myocardium the branches break up into fine fibres, called the Purkinje fibres.
- The AV bundle, bundle branches and Purkinje fibers convey electrical impulses from *the AV node to the apex of the myocardium where the wave of ventricular contraction begins*, then sweeps upwards and outwards, pumping blood into the pulmonary artery and the aorta.
- *The following figure shown the atrial and ventricular excitation*



The Heart Sounds:

- Two sounds are normally heard through a stethoscope

The first heart sound

It is soft low pitched; the average duration is 0.12 second

Causes:

1- Valvular factor:

- a) Closure of the A.V valves
- b) Vibrationsof chorda tendinae

2- Muscular factor, muscular contraction and tension of the walls of the ventricles.

3- Vascular factor, distention of the aorta and pulmonary artery

The second heart sound:

- It is sharper, higher in pitch and shorter in duration. Its average duration is 0.08 second. It occurs during the isometric relaxation phase.

Causes:

Valvular factor:

- The sudden closure of the semilunar valves. The second heart sound is loud and sharp when the aortic or pulmonary artery pressure is elevated.
- The aortic component of the second heart sound is best heard at the second right intercostal space at parasternal line, while the pulmonary component is heard at the second left intercostal space near the sternum.

The third heart sound

The fourth heart sound.

The Heart Rate:

Definition: *the number of beats per minute*

- The normal heart rate under basal condition is ranged from 50 to 90 beat per minute.

Factors affecting heart rate: these factors include

1. Autonomic nervous system.

- *Parasympathetic* decrease the rate and force of the heart beat.
- *Sympathetic* stimulation *increases* the rate and force of the heart beat.

2. Circulating chemicals.

- Adrenaline and noradrenalin *increase* the heart rate.
- Thyroxin *increase* heart rate by their metabolic effect.

3. Position.

- When the person is upright, the heart rate is usually *faster than* when lying down.

4. Exercise.

- *increased* heart rate

5. Emotional states.

- During excitement, fear or anxiety the heart rate is *increased*.

6. Gender.

- The heart rate is *faster* in women than men.

7. Age.

- In babies and small children the heart rate is *more rapid than* in older children and adults.

8. *Temperature.*

- The heart rate *rises and falls* with body temperature.

9. *Baroreceptor reflex.*

- These are nerve endings sensitive to pressure changes (stretch) within the vessel, situated in the arch of the aorta and in the carotid sinuses.
- A rise in blood pressure in these arteries stimulates the baroreceptors, increasing their input to the CVC. The CVC responds by increasing parasympathetic nerve activity to the heart; this slows the heart down. At the same time, sympathetic stimulation to the blood vessels is inhibited, causing vasodilatation.
- The CVC responds by increasing sympathetic drive to the heart to speed it up.

The Arterial Blood Pressure

Definition:

The force exerted by the blood on the walls of the arteries.

Measurement of arterial blood pressure

- *Direct method*
- Indirect Measurement of the arterial blood pressure in man by sphygmomanometric method



Normal value of blood pressure:

- young adult individual at the level of heart at rest 120/80 mmHg .

ABP = COP X PR

A-Systolic blood pressure:

It is the maximal pressure during systole it is about 90 - 140 mmHg.

B-Diastolic blood pressure:

It is the minimal pressure just before ventricular systole it is about 60 - 90 mmHg.

Physiological factors affecting the normal value of blood pressure

1-Age

At birth 70-80/40-50 mm/Hg

At 20 years 120/80 mm/Hg

ABP becomes normally 150/90 mm/Hg **after the age of 60 years**

2-Sex is faster in women than men

3-Emotions During excitement, fear or anxiety the heart rate is **increased**

4-Constitution (ABP of obese people is usually above the normal)

5-Sleep Decreases the ABP by 15-30 mmHg if not accompanied by dreams.

6-Meals During digestion the ABP increases by nearly 5-10 mmHg

7-Exercise (increased)

8-Gravity: In the recumbent position. The ABP in the brachial artery increases by about 25 mmHg if the arms are hanged below the bed. ABP decreases if the arm is raised

Hypertension:

Definition:

Sustained elevation of systolic blood pressure more than 130mm Hg and diastolic blood pressure more than 90 mmHg for three successive records.

Hypotension – low BP in which systolic pressure is below 100 mm Hg

Cardiac Output (COP)

Stroke volume:- Volume of blood ejected from each ventricle per beat

Stroke volume (SV) = 70 ml

$$= EDV - ESV$$

EDV = 110-120 mL

ESV = 40-50 mL

EDV represents the amount of blood that collects in a ventricle during diastole. While **ESV** represents the volume of blood remaining in a ventricle after it has contracted.

Minute volume = Cardiac output (COP)

Volume of blood ejected from each ventricle per minute.

$$= SV \times HR = 5 \text{ liters / min}$$

$$= (120 - 50) \times 70$$

$$= 70 \times 70$$

$$= 4900 \text{ ml/min.}$$

≈ 5 L/min.

Cardiac index: *cardiac output/ square meter of body surface area*
= 3.2 liter / min / m²

◆ **Regulation of CO:**

■ CO regulation depends on the regulation of:

- a. Stroke volume, &
- b. Heart rate

- The following table summarizes factors affecting cardiac output

Box 5.1 Summary of factors affecting cardiac output

Cardiac output = Stroke volume × Heart rate

Factors affecting stroke volume:

- VEDV (ventricular end-diastolic volume)
- Venous return
 - position of the body
 - skeletal muscle pump
 - respiratory pump
- Strength of myocardial contraction
- Blood volume.

Factors affecting heart rate:

- Autonomic nerve stimulation
- Circulating chemicals
- Activity and exercise
- Emotional states
- Gender
- Age
- Body temperature
- Baroreceptor reflex.

2- The Vascular System:

Vascular Tree

- Closed system of vessels which consists of :

a- Arteries

- Carry blood away from heart to tissues

b- Arterioles

- Smaller branches of arteries

c- Capillaries

- Smaller branches of arterioles
- Smallest of vessels across which all exchanges are made with surrounding cells

d- Venules

- Formed when capillaries rejoin
- Return blood to heart

e- Veins

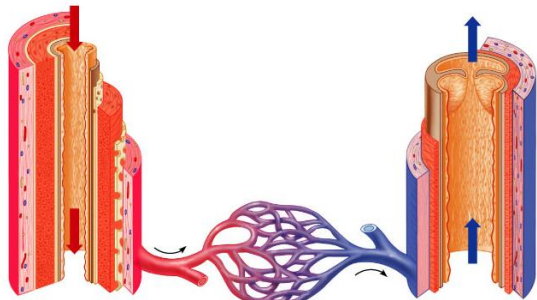
- Formed when venules merge
- Return blood to heart

NOTE: as shown in the right figure

- arteries & veins run in pairs
- veins have larger diameters
- veins have thinner walls
- veins are more compliant (or less resilient)

Arteries take blood
from the heart

Veins return
blood to the heart



Capillaries

1. Connect arteries & veins
2. Site where gasses, nutrients & wastes are exchanged between blood and tissues

16



3. Blood

The blood transports substances around the body through a large network of blood vessels.

The blood consists of two parts:

A. Plasma.

This is mainly water with a wide range of substances dissolved or suspended in it.
These include:

- Nutrients absorbed from the alimentary canal
- Oxygen absorbed from the lungs
- Chemical substances synthesized by body cells, e.g. hormones
- Waste materials produced by body cells to be eliminated from the body by excretion.

B. Blood Cells

- There are three groups, classified according to their functions
- 1. ***Erythrocytes (Red blood cells)***
 - Transport oxygen and carbon dioxide between the lungs and all body cells.
- 2. ***Leukocytes (White blood cells)***
 - protect the body against microbes
- 3. ***Thrombocytes (Platelets)***
 - Play an essential part in the process of blood clotting.